

Flag

S 0

Z 1

AC 0

P 1

C 0

tion

ex

Dec

ie

Load me at

1 LDA 2500H

2 MOV B,A ; B = First number

3

4 LDA 2501H

5 MOV C,A ; C = Second number

6

7 LOOP: MOV A,B

8 CMP C

9 JZ STORE

10

11 JC SWAP

12

13 SUB C

14 MOV B,A

15 JMP LOOP

16

17 SWAP: MOV A,C

18 MOV D,A

19

20 MOV A,B

21 MOV C,A

22

23 MOV A,D

24 MOV B,A

25

26 JMP LOOP

27

28 STORE: MOV A,B

29 STA 2502H

30

31 HLT

DataStackKeyPadMemor

Start2500

Address (Hex)	Address	Data
09C4	2500	2
09C5	2501	4
09C6	2502	2
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0
09D2	2514	0
09D3	2515	0

Line No	Assembler Message
0	Program assembled successfully